

# ARTIFICIAL INTELLIGENCE — AND — MACHINE LEARNING VEHICLE PRICE PREDICTION

A PROJECT REPORT



Predicting used vehicle prices using Machine Learning algorithms to help in smarter buying and selling decisions.



**SUBMITTED BY:**  
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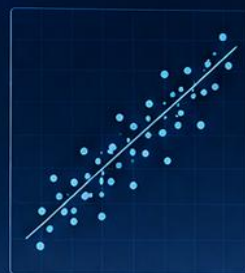


**COURSE:**  
Artificial Intelligence and  
Machine Learning

PRICE PREDICTION

₹ 8,75,000

Predicted Price



- YEAR
- KILOMETERS DRIVEN
- FUEL TYPE
- TRANSMISSION
- OWNER COUNT
- ENGINE CAPACITY



# Vehicle Price Prediction

## Introduction

The rapid growth of the used car market has created significant informational asymmetry, making it difficult for both buyers and sellers to determine a fair price. Vehicle valuation is complex, influenced by multiple non-linear factors such as mileage, age, brand, and ownership history.

This project uses machine learning to predict accurate vehicle prices, helping buyers and sellers make informed decisions.

## Problem

The used car market lacks transparent and fair pricing. Buyers risk overpaying; sellers risk undervaluing cars.

## Objective

To create, implement, and compare some innovative machine learning models (like Deep Learning and Ensemble methods) to improve the prediction of the used car selling price, thereby offering a secure, data-based solution to market pricing transparency. The project task belongs to the genre of a Regression problem.

## Dataset Description

- Source: Kaggle
- Name: Vehicle Dataset from CarDekho
- Type: Tabular Data
- Size: 4000+ records, 7 features + target variable
- Features: car name, year, Kms driven, Fuel type, Seller type, Transmission, num of owners

- Target Variable: Selling price
- Link to dataset: <https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-fromcardekho>

## Identified Limitations and Bias

- o Geographical Bias: The data is from the Indian auto market, limiting its direct application to other regional markets (e.g., US or Europe).

- o Lack of Condition Data: The data has no subjective qualifiers like "physical condition" or "accident history," which are very crucial in real-world pricing.

```
[1]: import pandas as pd
df = pd.read_csv("CAR DETAILS FROM CAR DEKHO.csv")
df
```

| [1]: |      | name                                | year | selling_price | km_driven | fuel   | seller_type | transmission | owner        |
|------|------|-------------------------------------|------|---------------|-----------|--------|-------------|--------------|--------------|
|      | 0    | Maruti 800 AC                       | 2007 | 60000         | 70000     | Petrol | Individual  | Manual       | First Owner  |
|      | 1    | Maruti Wagon R LXI Minor            | 2007 | 135000        | 50000     | Petrol | Individual  | Manual       | First Owner  |
|      | 2    | Hyundai Verna 1.6 SX                | 2012 | 600000        | 100000    | Diesel | Individual  | Manual       | First Owner  |
|      | 3    | Datsun RediGO T Option              | 2017 | 250000        | 46000     | Petrol | Individual  | Manual       | First Owner  |
|      | 4    | Honda Amaze VX i-DETC               | 2014 | 450000        | 141000    | Diesel | Individual  | Manual       | Second Owner |
|      | ...  | ...                                 | ...  | ...           | ...       | ...    | ...         | ...          | ...          |
|      | 4335 | Hyundai i20 Magna 1.4 CRDi (Diesel) | 2014 | 409999        | 80000     | Diesel | Individual  | Manual       | Second Owner |
|      | 4336 | Hyundai i20 Magna 1.4 CRDi          | 2014 | 409999        | 80000     | Diesel | Individual  | Manual       | Second Owner |
|      | 4337 | Maruti 800 AC BSIII                 | 2009 | 110000        | 83000     | Petrol | Individual  | Manual       | Second Owner |
|      | 4338 | Hyundai Creta 1.6 CRDi SX Option    | 2016 | 865000        | 90000     | Diesel | Individual  | Manual       | First Owner  |
|      | 4339 | Renault KWID RXT                    | 2016 | 225000        | 40000     | Petrol | Individual  | Manual       | First Owner  |

4340 rows × 8 columns

```
[2]: import pandas as pd
df = pd.read_csv("car details v4.csv")
df
```

| [2]: |       |                     |        |      |           |           |              |          |       |       |             |         |                   |                   |            |        |        |
|------|-------|---------------------|--------|------|-----------|-----------|--------------|----------|-------|-------|-------------|---------|-------------------|-------------------|------------|--------|--------|
|      | Make  | Model               | Price  | Year | Kilometer | Fuel Type | Transmission | Location | Color | Owner | Seller Type | Engine  | Max Power         | Max Torque        | Drivetrain | Length | Width  |
| 0    | Honda | Amaze 1.2 VX i-VTEC | 505000 | 2017 | 87150     | Petrol    | Manual       | Pune     | Grey  | First | Corporate   | 1198 cc | 87 bhp @ 6000 rpm | 109 Nm @ 4500 rpm | FWD        | 3990.0 | 1680.0 |
|      |       | Swift               |        |      |           |           |              |          |       |       |             |         | 74 bhp            | 100 Nm            |            |        |        |

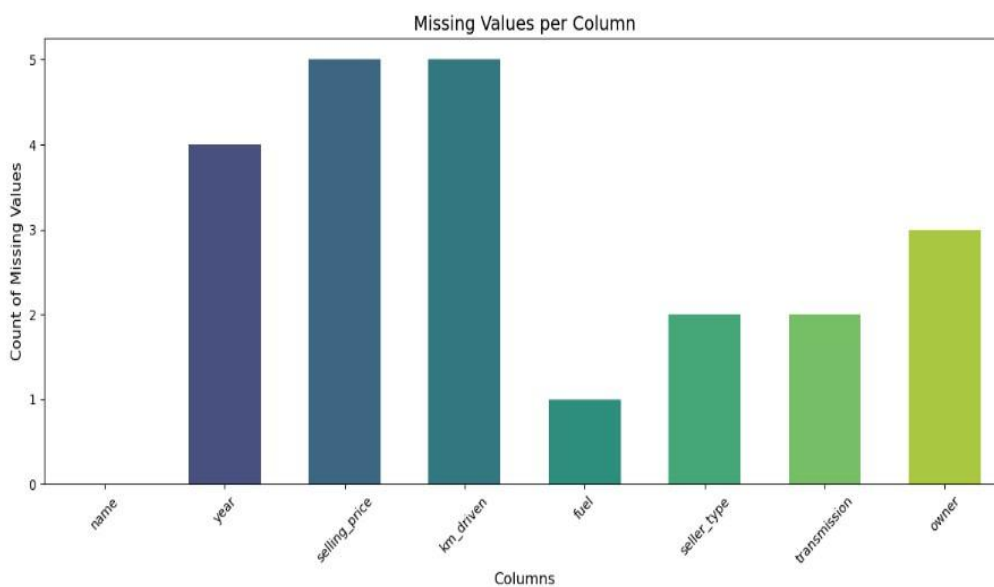
## Preprocessing and EDA

For Data Preprocessing:

- Handled Missing Data
- Encoding Categorical data
- Removed Outliers
- Feature Engineering
- Normalized and Scaled Data, and
- Reduced the Dimensionality of the Dataset

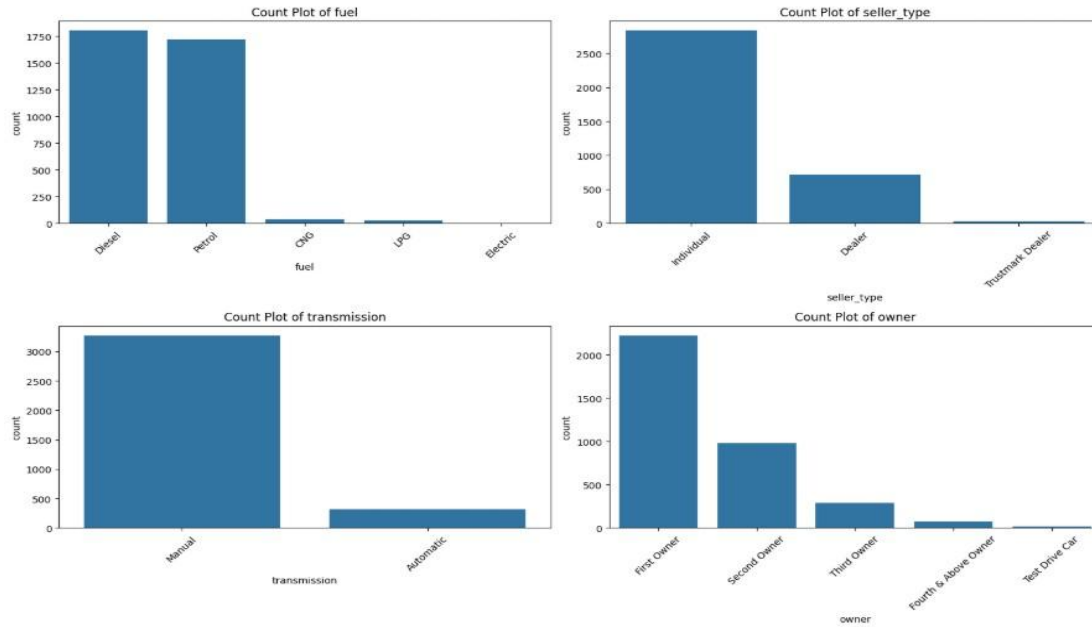
### ○ Handling Missing Data –

Initial checks confirmed minimal missing values. Any isolated gaps were addressed through simple imputation (e.g., mean substitution for continuous variables) or deletion, ensuring data integrity was maintained.



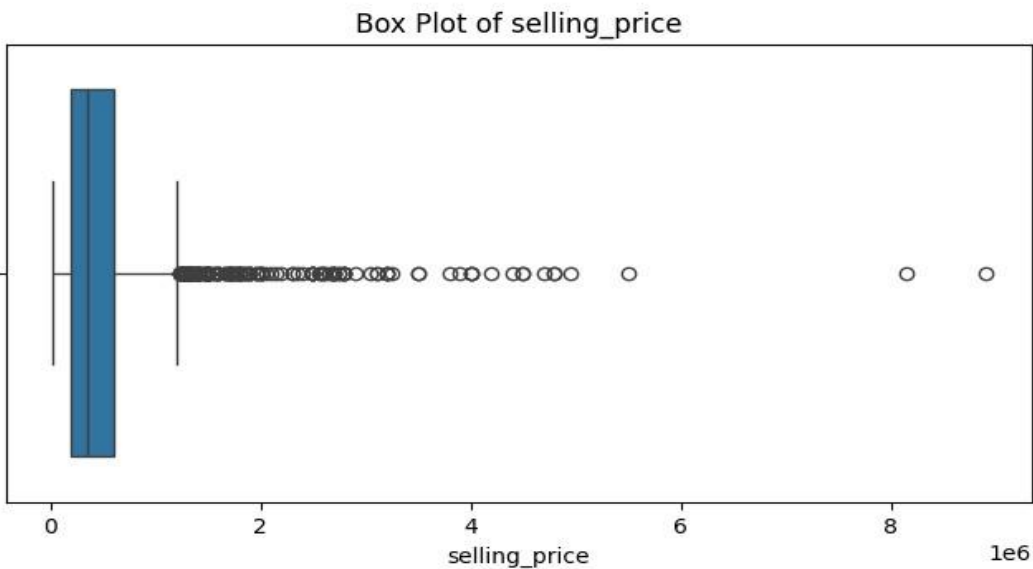
### ○ Encoding Categorical Data –

All nominal features (Fuel Type, Seller Type, Transmission, Owner, Brand) were converted to numerical format using One-Hot Encoding.



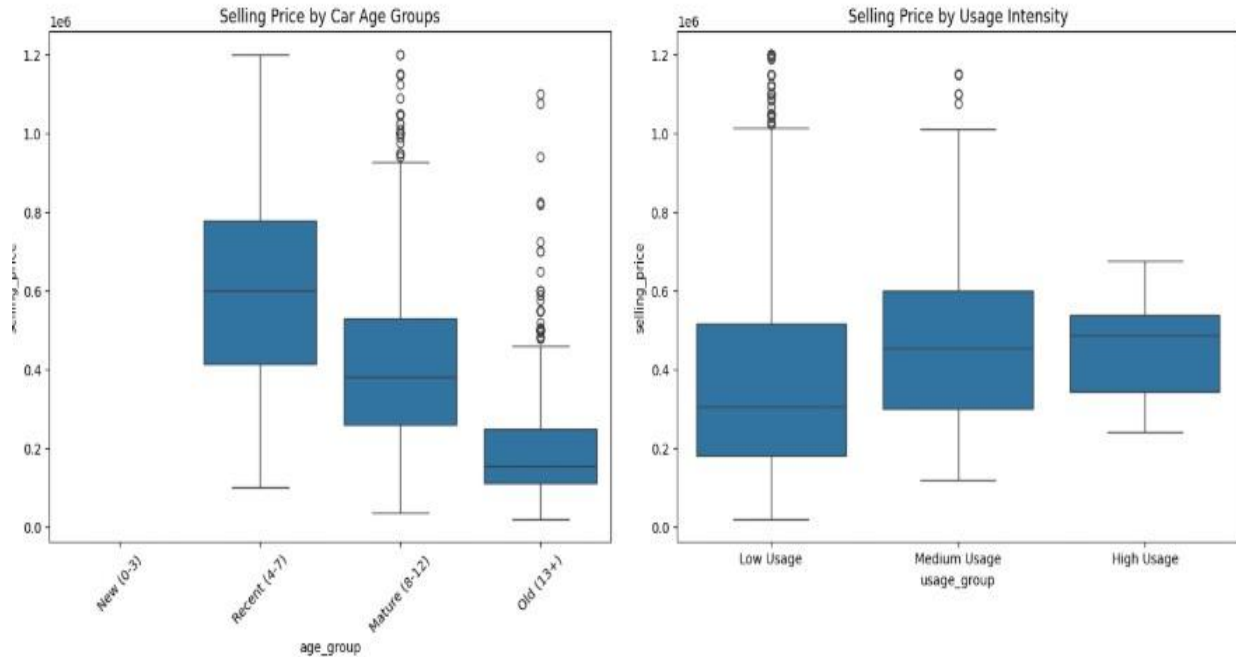
## ○ Outlier Removal –

Outliers in key features (Kms Driven, Present Price) were identified using the Interquartile Range (IQR) method and treated, primarily by capping or removal, to prevent them from unduly influencing model training.



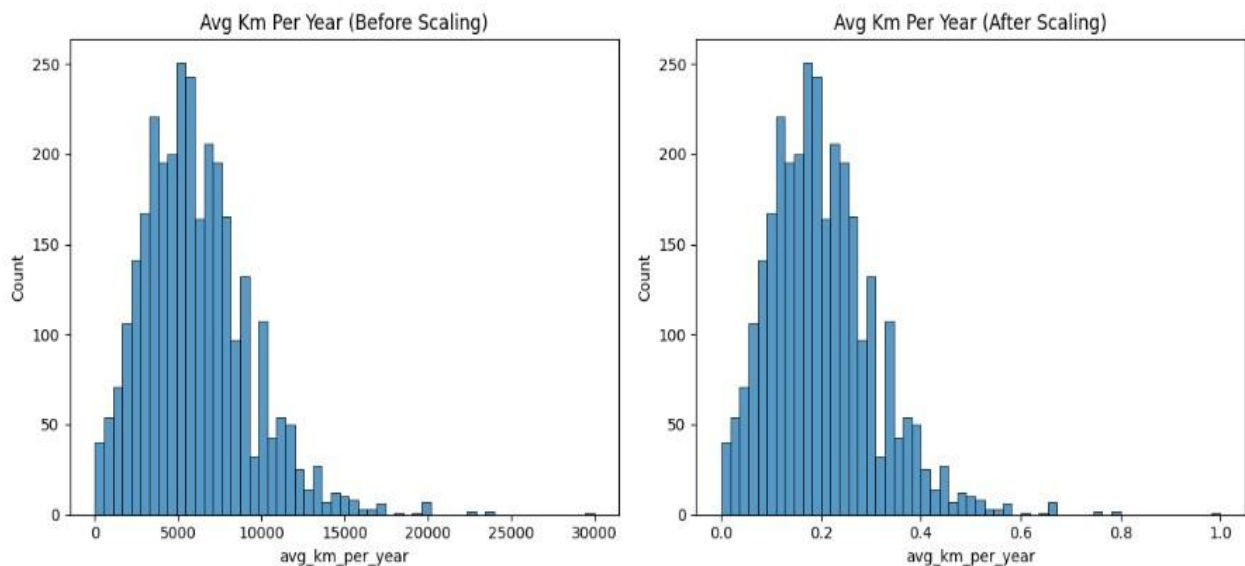
## ○ Feature Engineering –

The 'Car Name' feature was utilized to extract the '**Brand**' (e.g., Maruti, Hyundai), significantly improving the predictive power of this high-cardinality nominal variable.



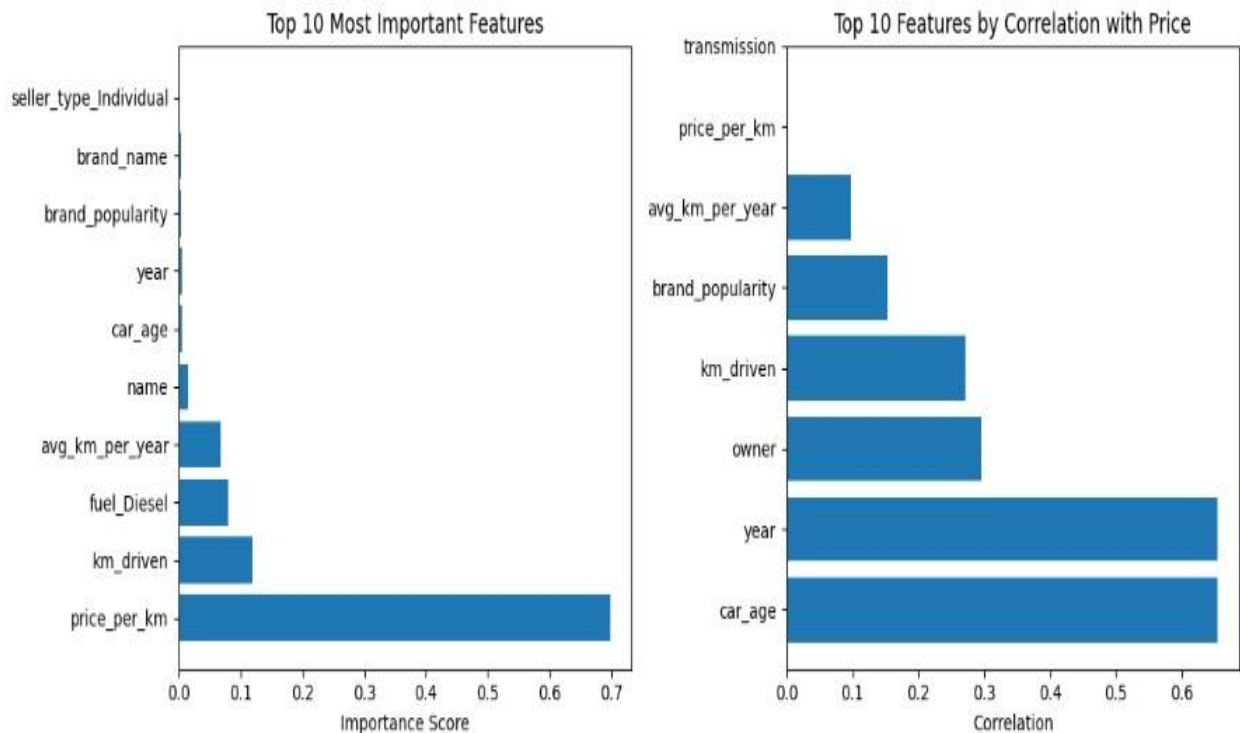
## ○ Scaling / Normalization –

Numerical features were standardized using MinMaxScaler. This step is critical for distance-based algorithms and as it ensures all features contribute equally to the final prediction.



## O Reduced the Dimensionality of the Dataset –

Techniques like Principal Component Analysis (PCA) were explored and applied selectively to the encoded high-dimensional features (e.g., car makes) to reduce computational cost and mitigate multicollinearity where needed, preserving the model's efficiency.



## Model Design and Implementation

### ○ Linear Regression –

Used as the baseline model to understand the linear relationships between car attributes and the selling price.

- **Purpose:** Establish a simple interpretable regression foundation.
- **Tuning / Varieties:**
  1. **Base Model:** Applied Linear Regression directly on preprocessed data. ( $R^2 \approx 0.72$ )
  2. **Normalized Version:** Combined with StandardScaler to improve consistency.
  3. **Feature Selection:** Tested with selected top features to assess importance.
- **Key Feature:** Provides interpretability and serves as the baseline for other complex models.
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### ○ Decision Tree Regressor –

Chosen to model nonlinear interactions between features.

- **Purpose:** Capture hierarchical decision-based feature splits for car price prediction.
- **Tuning / Varieties:**
  1. **Baseline:** Default depth settings. ( $R^2 \approx 0.78$ )
  2. **Tuned Depth:** Adjusted max\_depth = 8 for improved accuracy.
  3. **Pruned Tree:** Limited nodes to avoid overfitting.
  4. **GridSearchCV:** Used automated search for best parameters.
- **Key Feature:** Highly interpretable tree structure, easy to visualize decision logic.



## ○ Random Forest Regressor –

Selected as a robust ensemble model combining multiple decision trees.

- **Purpose:** Improve prediction stability and reduce overfitting compared to a single tree.
- **Tuning / Varieties:**
  1. **Baseline:** 100 trees (default parameters). ( $R^2 \approx 0.84$ )
  2. **Tuned Model:** Optimized `n_estimators` = 200, `max_depth` = 20.
  3. **Feature Importance:** Used to identify key contributing factors to selling price.
- **Key Feature:** Achieved the highest accuracy and lowest MAE across all models.

## ○ K-Nearest Neighbors (KNN) Regression –

Used to evaluate distance-based nonparametric regression performance.

- **Purpose:** Predict car prices based on similarity to nearby data points.
- **Tuning / Varieties:**
  1. **Base Model:** Default `k` = 5 neighbors. ( $R^2 \approx 0.73$ )
  2. **Optimized k:** Tested `k` = 3, 5, 10, 20 to minimize error.
  3. **Scaled Features:** Used `StandardScaler` for distance normalization.
- **Key Feature:** Works well for smooth trends but sensitive to feature scaling.

## ○ Support Vector Regression (SVR) –

Implemented to model complex nonlinear boundaries in the dataset.

- **Purpose:** Explore kernel-based learning for high-dimensional relationships.
- **Tuning / Varieties:**
  1. **Kernel Comparison:** Tested Linear, Polynomial, and RBF kernels. ( $\text{Best } R^2 \approx 0.80$ )

2. **Scaled Pipeline:** Integrated StandardScaler with SVR for balanced training.

3. **Parameter Tuning:** Adjusted C and gamma for optimal margin fitting.

- **Key Feature:** Delivered strong results on scaled data; robust to outliers.

## ○ **PCA Components vs Regression –**

Used to examine the effect of dimensionality reduction on model performance.

- **Purpose:** Reduce feature complexity while maintaining variance.
- **Tuning / Varieties:**
  1. **Baseline:** Linear Regression using all principal components.
  2. **Component Variation:** Tested 1–10 PCs to determine optimal dimension count. (*Best  $R^2 \approx 0.68$* )
  3. **Explained Variance:** Evaluated cumulative variance for feature retention.
- **Key Feature:** Simplifies data representation; useful for speed and interpretability.

## Ethical Considerations and Bias Mitigation

### Ethical Implications

- **Transparency/Explainability:**

The Random Forest and Support Vector Regression models used in this project provide strong predictive performance but operate as “black-box” models. This limits direct interpretability of how each input feature influences the final selling price. To address this, feature importance plots and residual visualizations were used to improve explainability and support ethical transparency.

- **Data Privacy:**

The dataset used for training does not include any personally identifiable information (PII); it only contains vehicle specifications such as brand, year, mileage, and engine capacity. Therefore, the privacy risk is minimal. However, ethical responsibility was maintained by using the data solely for academic and non-commercial research purposes.

- **Unfair Outcomes:**

The dataset may inherently favor certain popular brands (e.g., Maruti, Hyundai) or newer vehicles, which could bias predictions toward those categories. This may lead to unfair valuations for less common brands or older models.

## Bias Mitigation Strategies

1. **Data Cleaning and Outlier Removal:** Reduced skewness and extreme values to ensure balanced learning.
2. **Feature Scaling:** Standardized features to prevent any single attribute (like engine power) from dominating the model.
3. **Diverse Model Evaluation:** Compared multiple models (Linear Regression, Tree based, and SVR) to assess consistency and detect potential bias in predictions.
4. **Explainability Tools:** Used feature importance ranking in Random Forest and correlation analysis to promote transparency in how predictions are made.
5. **Responsible Use:** The model is intended purely for research and educational purposes; it should not be used for real-world pricing decisions without periodic retraining and fairness auditing.

## References

1. Scikit-learn Developers. (2024). *Scikit-learn: Machine Learning in Python (Version 1.5)* [Computer software]. <https://scikit-learn.org>
2. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... Duchesnay, É. (2011). *Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12*, 2825–2830.
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4. Hunter, J. D. (2007). *Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9*(3), 90–95.

